Linux Networking Overview

Advanced Embedded Software Development

with **Dan Walkes**



University of Colorado **Boulder**

Learning objectives:

Understand tips for working with IP based networking on Linux and Buildroot.

Linux Networking ifconfig/ip

- ifconfig was the traditional way to manage networking
- ip is the new command to manage networking
- ip addr - list ip address details

```
root@qemuarm64:~# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:12:34:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.7.2/24 brd 192.168.7.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::5054:ff:fe12:3402/64 scope link
       valid_lft forever preferred_lft forever
3: sit0@NONE: <NOARP> mtu 1480 qdisc noop qlen 1000
    link/sit 0.0.0.0 brd 0.0.0.0
```

https://www.linux.com/tutorials/replacing-ifconfig-ip/

Linux Networking - ip addr

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```

- lo: loopback port
- eth0 : local ethernet (may see enp prefix on systemd systems)
- sit0: tunneling for IPv6 over IPv4
- wlanX: wireless lan port

https://www.freedesktop.org/wiki/Software/systemd/PredictableNetworkInterfaceNames/

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- UP means the link is up
  - May or may not mean you are connected to the Internet
- link/ether lists the MAC address
  - Global unique reference to this port
- inet gives the IPv4 address/mask - How is this assigned?

https://linux.die.net/man/8/ip

Linux Networking - Interfaces Config

```
# cat /etc/network/interfaces
# interface file auto-generated by buildroot

auto lo
iface lo inet loopback

auto eth0
iface eth0 inet dhcp
  pre-up /etc/network/nfs_check
  wait-delay 15
  hostname $(hostname)
```

```
    Target options  --->
    Build options  --->
    Toolchain  --->
    System configuration  --->
    Kernel  --->
    Target packages  --->
    Filesystem images  --->
    /bin/sh (busybox' default shell)  --->
[*] Run a getty (login prompt) after boot  --->
[*] remount root filesystem read-write during boot
(eth0) Network interface to configure through DHCP
(/bin:/sbin:/usr/bin:/usr/sbin) Set the system's default PATH
[*] Purge unwanted locales
(C en_US) Locales to keep
    *** NLS support needs a toolchain w/ wchar, dynamic library ***
```

- For more complex configurations, update this file
  - Use a rootfs overlay on Buildroot
  - Use ROOTFS_POSTPROCESS_COMMAND on Yocto

http://manpages.org/etc-network-interfaces/5
https://patchwork.ozlabs.org/patch/458649/
https://www.yoctoproject.org/docs/latest/ref-manual/ref-manual.html#var-ROOTFS_POSTPROCESS_COMMAND